PRESIDENT'S OFFICE
REGIONALA ADMINISTRATION AND LOCAL GOVERNMENT
ARUSHA REGIONAL
FORM THREE TERMINAL EXAMINATION
PHYSICS

TIME: 3HOURS MAY, 2025

INSTRUCTIONS:

- This paper consists of sections **A, B** and **C** with a total of **eleven (11)** questions.
- Answer **ALL** questions in sections **A** and **B** and **two (2)** questions from section **C** in your answer sheet/booklet provided.
- Section A carries **sixteen (16)** marks, section **B fifty-four (54)** and section **C** carries **thirty (30)** marks.
- Cellular phones and any unauthorized materials are not allowed in the examination room.
- Non-programmable calculators and mathematical tables may be used.
- Where necessary, the following constants may be used: -
  i) Acceleration due to gravity, g=10N/kg
  ii) Density of sea water = 1025kg/m^3
  iii) Density of mercury = 13600kg/m^3
  iv) Density of water =1000kg/m^3
  v) Density of air = 1.3kg/m^3
  vi) 1 atm = 1x10^5pa
  vii) Pie (π) = 3.14

SECTION A: (16 Marks)

Answer **ALL** questions in this section

1. For each of the multiple-choice items (i) – (x), choose the correct answer from among of the given alternatives and write its letter in the box besides it.

   i) A student measured the diameter of a wire using a screw gauge and he repeated the experiment four times and listed the measurements. The correct measurement was:
   A 0.3cm C. 0.333cm
   B. 0.33cm D. 0.3333cm

   ii) Suppose you were given several physical quantities and you were asked to select the fundamental quantities. What would be your selection form the following:
   A. Length, weight and time B. length, mass and temperature
   C. mass, time and volume D. Length, density and time

iii) If 0.35 litres of water is mixed with 0.15 litres of mercury, then the density of the mixture will be:
A. 14.6g/cm^3 B. 7.3g/cm^3 C. 14.8g/cm^3 D. 4.78g/cm^3

iv) Why can water striders and pond skaters walk on water without sinking?
A. Because of process of osmosis
B. Because of property of capillarity
C. Because of presence of impurities in water
D. Because of the force of surface tension

v) Light rays travelling from a water medium to denser flint glass medium. What is the best student to describe the behaviour of light bending in dense flint glass medium?
A. It will bend towards the normal
B. It will bend away from the normal
C. No bending since it will reflect
D. Total internal reflection will occur

vi) Fatuma was cooking ugali in a good conducting container but she seems to use iron handle covered with plastic at its holding point to hold a cooking container. Why did she use plastic covered handle and not bare handle?
A. Plastic reflects heat back to the container
B. Plastic is a good absorber of hear energy
C. Plastic is a good emitted of heat energy
D. Plastic is a good insulator.

vii) A stone is thrown up from the top of a building of 45m, high at a speed of 12m/s. How long does it take to reach the ground?
A. 1.2 seconds B. 4.4 seconds C. 3.2 seconds D. 2.2 seconds

viii) A man pulls a square box of mass 10kg with the force of 50N and the box start to accelerate in the direction of force with 2m/s^2. What will be the value of the coefficient of friction and the type of friction force?

ix) Which statement is true about a body whose work done is said to be zero?
A. Its displacement is in the opposite direction to the force applied
B. Its displacement is in a direction perpendicular to the force applied
C. Its displacement is in the same direction to the force applied
D. Its displacement is at an angle to the direction of the applied force

x) What makes a coin inside the water at the bottom of a vessel appears to rise when viewed from above?
A. Reflection of light B. refraction of light
C. Diffraction of light D. Mirage of light.

2. Match the descriptions of the terms used in light in **list A** with its corresponding concept used in light in **list B** by writing the letter of the correct response beside the corresponding its number in the answer sheet/booklet provided.

LIST A	LIST B
i) The centre of the curved mirror	A. Centre of curvature
ii) The centre of the glass sphere	B. Concave
iii) The distance between the centre of curvature and the pole	C. Convex
iv) The distance between the centre of curvature and the principal focus	D. Focal length
v) The line joining the centre of curvature and the pole	E. pole
vi) The mid-point between the centre of curvature and the pole	F. Polished surface
	G. Principal axis
	H. Principal focus
	I. Radius of curvature

SECTION B: (54 Marks)

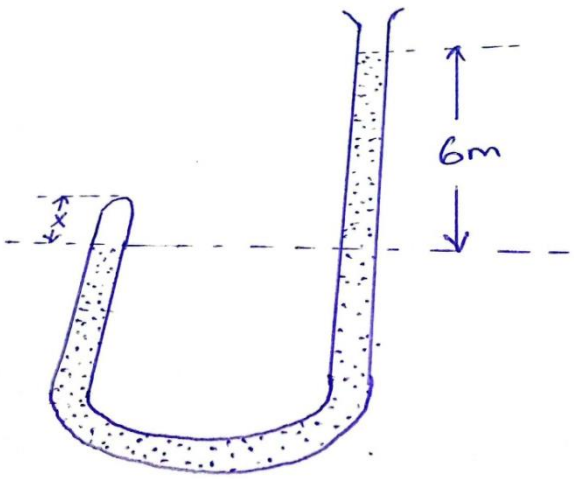
Answer **ALL** questions from this section

3. a) With aid of labelled ray diagram, show how you can obtain an image of an object at a very long distance on a converging mirror. **(05 marks)**

   b) A 14 – years old student is not able to see clearly the questions written on the blackboard placed at a distance of 5m from him.
      i) Name the defect of vision, he is suffering from **(02 marks)**
      ii) Briefly, explain how this defect can be corrected. **(02 marks)**

4. a) i) A tall vertical cylinder filled with water is kept on a horizontal table. Two small hole A and B are made on the wall of the cylinder. A near the middle and B, just below the free surface of water. State and explain your observation. **(02 marks)**

      ii) A glass tube is open at one end and seated at the other end. A dense liquid of density 14000kg/m^3 is poured into the tube. The liquid traps air in the sealed end as shown in the diagram below. Calculate the pressure of the trapped air.



b) A uniform half metre rule is freely pivoted at the 15cm mark and it balances horizontally when a body of mass 80g is hung from the x cm mark on a rule. If y is the distance in centimetres from the pivot to the x cm mark, prove that, the mass of rule in grams is given by: m = 8y **(04 marks).**

5. a) i) Imagine you are standing in a bus that suddenly turns sharply. Why do you feel like you are being pushed sideways, even though nothing is pushing you directly?**(02 marks)**

ii) A 2kg object travelling at 3m/s to the right collides head-on with a 3kg object travelling at 4m/s to the opposite direction. What are the velocities of the two objects after collision, if the collision is perfectly elastic. (**03 marks)**

b) A pulley system is made up of 8 pulleys. An effort of 200N is applied on the system of 80% efficiency. Draw a pulley system and then determine the maximum load that can be raised by the effort. **(04 marks)**

6. a) The temperature of the surface of the sun is approximately 77k. The form two students were asked to read this temperature on the Celsius and Fahrenheit scale but they failed to do so. Help them **(05 marks)**

b) Suppose you are given a bar magnet whose poles are not labelled and your friends need to know the north pole. As an expert, explain how you could help them in determination that, the end of the bar is the north pole in the laboratory. **(04 marks)**

7. a) i) A form three students performed an experiment on determination of the acceleration due to gravity by using a pendulum bob. Briefly describe the energy changes when the bob of a simple pendulum swings from one side to another. **(03 marks)**

ii) From a (i) above, explain why a pendulum did not return to a point, higher than the point from which it was released? **(02 marks)**

b) A mass 3kg hangs at the end of a sting. Find the horizontal force needed to pull the mass sideway until the string is at 30° to the vertical, also find the tension in the string at the final position. **(04 marks)**

8. a) Your physics teacher has assigned you to measure themass of stone using a triple beam balance. What are the steps that you would follow while you are accomplishing your task assigned by your physics teacher? **(05 marks)**

b) A cargo ship has a cross sectional area of $100m^2$. If it is loaded with 36 empty containers each of 2.5625 tonnes mass, how deep would the ship sink in the sea water? **(04 marks)**

SECTION C (30 Marks)

Answer any **two (02)** questions from this section

9. a) Small pieces of tissue are attracted to a charged comb. Soon after sticking to the comb, the pieces of tissue are repelled from it. Explain. **(07 marks)**

   b) You are provided with three resistors of resistances 4Ω, 3Ω and 2Ω respectively.
      i) What is the minimum resistance which can be obtained by connecting the three resistors?**(05 marks)**

      iii) Sketch the diagram showing the connection in 9 (b) (i) above **(03 marks)**

10. a) Under what condition applied force is said to be less than friction force? **(05 marks)**

   b) A block of mass m slides down a plane inclined at an angle of 60° to the horizontal with an acceleration equal to half of the acceleration due to gravity. Show that the coefficient of kinetic friction, $\mu_k = \sqrt{3} - 1$ **(05 marks)**

   c) Form three students at a certain secondary school, are claiming that friction is useless. As a form three student from Changanyikeni Secondary school, prove them wrong in five (5) points. **(05 marks)**

11. a) With example, explain why some physical quantities have both magnitude and direction but not vectors. **(03 marks).**

   b) A motor boar travelling 6m/s, east encounters a current travelling 3.7m/s, South. Calculate: -
      i) The resultant velocity of the motor boat **(02½ marks)**
      ii) The time it takes the boat to travel shore to shore, if the width of the river is 120 meters wide. **(02 marks).**
      iii) The distance downstream does a boat reach the opposite shore. **(01½ marks)**

   c) i) Distinguish between absolute velocity and relative velocity. **(02 marks)**

      ii) The air plane usually encounters winds blowing with respect to an observer on the ground. By using sketched diagram, identify these winds and their effects on the airplane. **(04 marks)**